

Intelligent Crop Yield Prediction and Agricultural Risk Classification System Using Machine Learning

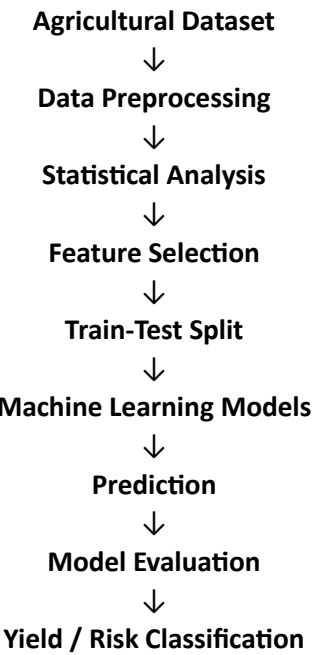
1. Proposed System

Input Dataset

Feature	Example
Rainfall (mm)	820
Temperature (°C)	27.5
Humidity (%)	68
Soil pH	6.5
Fertilizer Usage (kg/ha)	120
Pesticide Usage (kg/ha)	18
Cultivated Area (ha)	2.5
Crop Type	Rice
Soil Type	Loamy
Season	Kharif
Yield (tons/ha)	4.2
Risk Level	Low

2. Machine Learning Workflow

Workflow



3. Models to Implement

Decision Tree

Used for **risk classification**.

Classes:

-  Low Risk
-  Moderate Risk
-  High Risk

Random Forest

Neural Network

Genetic Algorithm

It can select the most important features:

Rainfall, Temperature, Humidity, Soil pH, Fertilizer Usage, Pesticide Usage, Cultivated Area

4. Data Processing Using Pandas

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv("agriculture_data.csv")
```

```
df = df.drop_duplicates()
```

```
df['Rainfall'] = pd.to_numeric(df['Rainfall'], errors='coerce')
```

```
df['Temperature'] = pd.to_numeric(df['Temperature'], errors='coerce')
```

```
df['Rainfall'] = df['Rainfall'].fillna(df['Rainfall'].median())
```

```
df['Temperature'] = df['Temperature'].fillna(df['Temperature'].median())
```

```
print(df.describe())
```

```
result = df.groupby('Crop_Type')['Yield'].agg(['mean', 'max', 'min'])
```

```
result = result.sort_values('mean', ascending=False)
```

```
print(result)
```

This demonstrates:

- DataFrame operations
- Filtering
- Missing-value handling

- Grouping
- Sorting
- Statistical analysis

5. Decision Tree

Your Decision Tree can classify agricultural risk.

Example:

Input

Rainfall = 850 mm

Temperature = 26°C

Humidity = 70%

Soil pH = 6.4

Fertilizer = 120 kg/ha

Output

Predicted Risk: LOW RISK

Another example:

Rainfall = 320 mm

Temperature = 38°C

Humidity = 35%

Soil pH = 8.1

Output:

Predicted Risk: HIGH RISK

6. Sample Model Results

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	84.6%	83.9%	84.2%	84.0%
Random Forest	91.3%	90.8%	91.1%	90.9%
Neural Network	89.7%	89.2%	88.9%	89.0%

Conclusion can be:

Random Forest achieved the highest classification performance and was selected as the most suitable model for agricultural risk classification.

7. Sample Prediction Output

Parameter	Input
Rainfall	780 mm
Temperature	27.2 °C
Humidity	72%
Soil pH	6.6
Fertilizer	110 kg/ha
Pesticide	15 kg/ha
Cultivated Area	2.8 ha
Crop Type	Rice
Season	Kharif

Prediction

Predicted Yield: 4.35 tons/ha

Agricultural Risk: ● **LOW RISK**

Recommended Action:

Maintain current irrigation and fertilizer levels.

8. Output

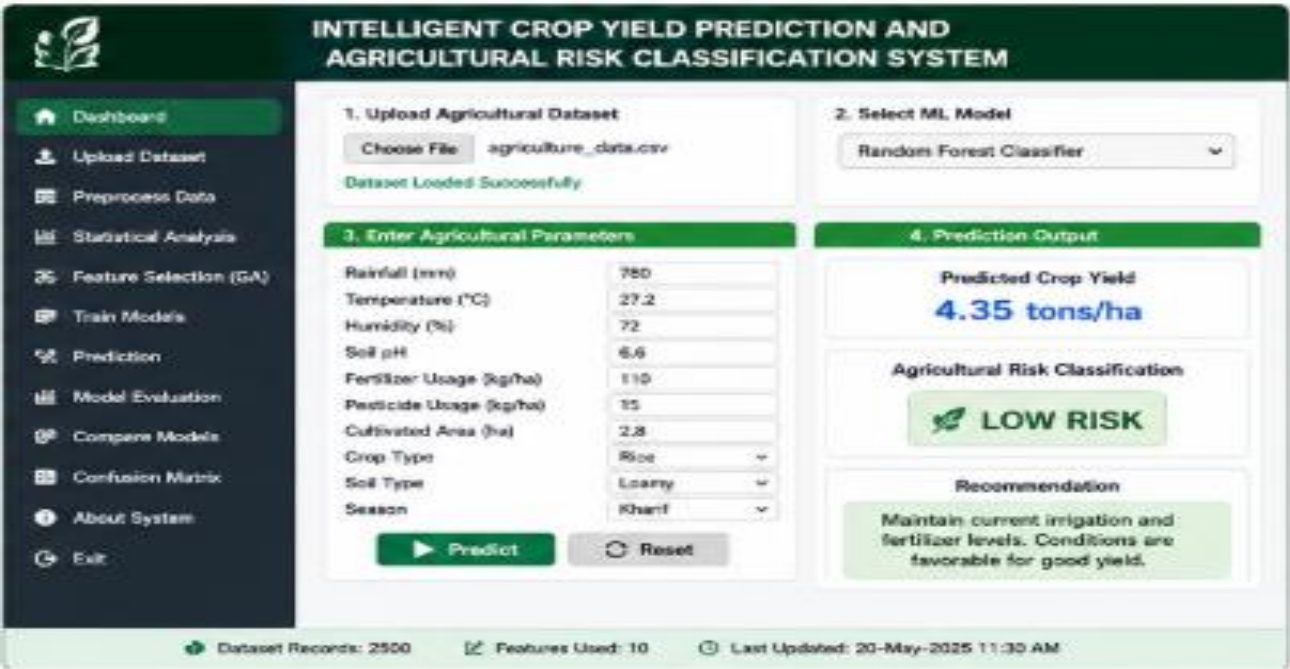


Figure 1: Intelligent Crop Yield Prediction and Agricultural Risk Classification System Interface

Figure 2 – Prediction Output



Figure 2: Crop Yield Prediction and Risk Classification Output

Figure 3 – Model Evaluation



Figure 3: Machine Learning Model Performance and Confusion Matrix

These three figures will make your PPT look much more like a **complete implemented project**, rather than only theoretical work.

9. Pseudocode

START

Load agricultural dataset

Display dataset information

Check missing values and duplicate records

Clean missing and invalid values

Convert categorical data into numerical form

Perform statistical analysis

Select important features
Split dataset into training and testing sets
Train Decision Tree model
Train Random Forest model
Train Neural Network model
Evaluate all models
Compare accuracy, precision, recall and F1-score
Accept new agricultural parameters
Predict crop yield
Classify agricultural risk
Display prediction and recommendation
Generate graphs and confusion matrix
Save processed results
STOP

10. SDG Connection

For the highlighted **Reflection** section, connect your project to:

SDG 2 – Zero Hunger

Crop yield prediction can help improve agricultural productivity and support food security.

SDG 9 – Industry, Innovation and Infrastructure

The system uses machine learning and data-driven technology for smart agriculture.

SDG 12 – Responsible Consumption and Production

Risk prediction can help farmers use fertilizer, pesticides and other resources more efficiently.

SDG 13 – Climate Action

Rainfall and temperature analysis can help identify agricultural risks caused by changing environmental conditions.